Mohammad Elayan, MSc

Graduate Research & Teaching Assistant

Dept. of Civil & Environmental Engineering University of Nebraska-Lincoln, Lincoln, NE, USA

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Profile

PhD student of the Civil & Environmental Engineering at the University of Nebraska–Lincoln. Research focuses on intelligent, human-aware autonomous driving in mixed traffic, with an emphasis on balancing competing driving objectives and modeling driver heterogeneity. Nearly a decade of transportation engineering experience in Dubai, leading numerous projects across the MENA region.

Education

PhD, Civil Engineering (Transportation Engineering) — 2023–Present
University of Nebraska-Lincoln, Lincoln, NE, USA — Ongoing, GPA 4.000

MSc, Civil Engineering (Transportation Engineering) — 2011–2013
Jordan University of Science & Technology, Irbid, Jordan — 91.6% · Rank 1

BSc, Civil Engineering (Transportation Engineering) — 2006–2011
Jordan University of Science & Technology, Irbid, Jordan — 82.3% · Rank 16

Research

Peer-Reviewed Publications 8

1. **Elayan, M.**, & Kontar, W. (2026). "Can Automated Vehicles Have it All? A Consensus Framework for Diagnosing Behavioral Trade-Offs in Mixed Traffic." *Transportation Research Part C: Emerging Technologies*. [Link]
2. **Elayan, M.**, & Kontar, W. (2026). "Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach." *2026 IEEE Intelligent Vehicles Symposium (IV)*, Plymouth, MI. [Link]
3. **Elayan, M.**, & Kontar, W. (2026). "Consensus-Aware AV Behavior: Trade-Offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, Gold Coast, Australia. [Link]
4. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). "Better Safety Analyses through Smarter Data: Adding Open-Street-View and Traffic Calibrated-LBS Data to Pedestrian Crash Analysis in Lincoln, NE." *Transportation Research Record*. [Link]
5. **Elayan, M.**, Aldridge, N., Hawkins, J., & Nam, Y. (2025). "Integrating StreetLight, EPS Smart Location Data and Road Attributes: A Random Forest Approach to Multi-Modal Traffic Calibration in Lincoln, Nebraska." *Journal of Transportation Engineering, Part A: Systems*, Vol. 151(8). [Link]
6. Al-Khateeb, G., Al-Suleiman, T., Khedaywi, T., & **Elayan, M.** (2016). "Studying Rutting Performance of Superpave Mixtures." *Road Materials and Pavement Design*, Vol. 19(2). [Link]
7. Alsheyab, M., Khedaywi, T., & **Elayan, M.** (2013). "Laboratory Study on Solidification/Stabilization of Unwanted Medications Using Asphalt as a Binder." *Journal of Material Cycles and Waste Management*, Vol. 15(2). [Link]
8. Al-Suleiman, T., & **Elayan, M.** (2013). "Gap Acceptance Behavior at U-turn Median Openings: Case Study in Jordan." *Jordan Journal of Civil Engineering*, Vol. 7(3). [Link]

Conference Papers & Presentations 12

1. **Elayan, M.** (2027). "Driving with Strangers: Behavioral Consensus and Driver Heterogeneity for Autonomous Vehicles in Mixed Traffic." *TRB 106th Annual Meeting, Emerging PhD Research in Traffic Flow Theory, Operations, and ITS (ACF00)*, Washington, DC. [Accepted]
2. Gabrys, G., **Elayan, M.**, Armantalab, O., & Kontar, W. (2027). "Where Pharmaceutical Deserts Are and How to Serve Them: A Geospatial Index and Autonomous Routing Study in Rural Nebraska." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
3. **Elayan, M.**, & Kontar, W. (2027). "Learning Behavioral Consensus for Autonomous Driving in Mixed Traffic." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
4. **Elayan, M.**, Armantalab, O., & Kontar, W. (2027). "Quantum-Inspired Modeling of Driving Behavior." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
5. **Elayan, M.**, & Kontar, W. (2026). "Behavioral Heterogeneity as Quantum-Inspired Representation." *IEEE ITSC*, Naples, Italy. [Presented]
6. **Elayan, M.**, & Kontar, W. (2026). "Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach." *IEEE Intelligent Vehicles Symposium (IV)*, Detroit, MI. [Presented]
7. **Elayan, M.**, & Kontar, W. (2026). "Consensus-Aware AV Behavior: Empirical Pareto-Guided Reinforcement Learning for Mixed Traffic." *NSF NAIRR Annual Meeting*, Arlington, VA. [Presented]
8. **Elayan, M.**, & Kontar, W. (2026). "The Empirical Pareto Frontier of Automated Driving." *College of Engineering 7th Annual Graduate Student Symposium*, UNL. [Presented]
9. **Elayan, M.**, & Kontar, W. (2026). "A Unified Framework for Consensus-Aware AV Behavior." *TRB 105th Annual Meeting*, Washington, DC. [Presented]
10. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2025). "Multi-Stage Clustering of Daily Activity Time Series." *TRB 105th Annual Meeting*, Washington, DC. [Presented]
11. **Elayan, M.**, & Kontar, W. (2025). "Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *IEEE ITSC*, Gold Coast, Australia. [Presented]
12. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). "Better Safety Analyses through Smarter Data." *TRB 104th Annual Meeting*, Washington, DC. [Presented]

Reports and Preprints 5

1. Armantalab, O., **Elayan, M.**, Zhao, L., Huynh, N., & Kontar, W. (2026). "A Quantum-Inspired Representation of Dynamic Travel Behavior." [Link]
2. **Elayan, M.**, & Kontar, W. (2026). "Behavioral Heterogeneity as Quantum-Inspired Representation." *arXiv preprint arXiv:2603.22729*. [Link]
3. **Elayan, M.**, & Kontar, W. (2026). "Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach." *arXiv preprint arXiv:2601.18913*. [Link]
4. **Elayan, M.**, & Kontar, W. (2025). "Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *arXiv preprint arXiv:2505.04379*. [Link]
5. Nam, Y., Hawkins, J., Butler, D., Aldridge, N., **Elayan, M.**, & Yoo, J. (2024). "Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data." *Nebraska Dept. of Transportation, Technical Report No. SPR-FY23(025)*. [Link]

Under Review 3

1. **Elayan, M.**, Armantalab, O., & Kontar, W. (2026). "Quantum-Inspired Modeling of Driving Behavior." *Transportation Research Part B: Methodological (Special Issue: Mathematical Foundations for Trustworthy AI Applications in Transportation Systems)*.
2. Armantalab, O., **Elayan, M.**, Zhao, L., Huynh, N., & Kontar, W. (2026). "A Quantum-Inspired Representation of Dynamic Travel Behavior." *Transportation Research Part C: Emerging Technologies*.
3. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2026). "Multi-Stage Clustering of Daily Activity Time Series: Patterns and Socio-Demographic Insights." *Transportmetrica A: Transport Science*.

Research Projects

- *Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data.* 2022–2024
  Nebraska Department of Transportation, USA.
- *Introducing New Technologies and Advanced Methodologies for Asphalt Mixtures of Highway Pavements in Jordan.* 2008–2012
  Ministry of Higher Education and Scientific Research, Amman, Jordan.

Grants & Awards

- NSF National AI Research Resource (NAIRR) Pilot Grant 2025
  "Consensus-Seeking Autonomous Agents in Complex Cyber-Physical Transportation Systems" (Award No. NAIRR250132). National Science Foundation, USA. *Role: Researcher.*

Fellowships & Scholarships

- University of Nebraska-Lincoln College of Engineering Professional Development Fellowship 2025, 2026
- Patrick T. McCoy Fund for Engineering Excellence 2025, 2026
- MATC & NTC Travel Fund Scholarship 2024, 2025, 2026

Experience

Graduate Research & Teaching Assistant Jan 2023 – Present
University of Nebraska-Lincoln, Lincoln, NE, USA

Member of Dr. Wissam Kontar's research team at the Civil & Environmental Engineering Department and Nebraska Transportation Center. Research focuses on traffic agent heterogeneity and automated vehicle behavior control.

Teaching Assistantships:

- CIVE 202, Civil Engineering Analysis II (programming in civil engineering)
- CIVE 361, Principles of Transportation Engineering

Transportation Engineer / Senior Transportation Engineer Jun 2013 – Jan 2023
TrafQuest, Dubai, United Arab Emirates

Contributed to **100+ projects** across the UAE and MENA, serving as Project Manager for the majority, including strategic developments generating up to 26,000 peak-hour trips.

Core roles:

- **Traffic operations:** analyzed roadway network performance, identified deficiencies, and developed feasible congestion-reduction solutions.
- **Modeling and simulation:** built microsimulation and travel demand models to forecast trip generation and evaluate future scenarios.
- **Multimodal integration:** evaluated connectivity across metro, bus, marine, taxi, pedestrian, and cycling modes, prioritizing clean and active transport.
- **Data analysis:** processed and interpreted large datasets, including parking demand analyses of 500,000+ records at 15-minute resolution.
- **Safety and design:** identified safety issues, conducted concept design reviews, and developed access, drop-off, and wayfinding designs.
- **Technical authoring:** authored engineering manuals [NAME/SCOPE] and prepared technical reports for clients and authorities.

Project category	*Total*	*Engineer*	*Manager*
Transportation Master Plan	21	13	8
Traffic Impact Study	65	25	40
Transportation Integration Study	11	1	10
Strategic Freight Transport Study	7	7	0
Parking Study	3	1	2
Traffic Management Plan	3	1	2
Evacuation Plan	1	0	1
Engineering Manual	1	1	0
Total	**111**	**48**	**63**

Graduate Research & Teaching Assistant 2011 – 2013
Jordan University of Science and Technology, Irbid, Jordan

Research in traffic operations, highway materials, and waste management: data collection, laboratory experiments, statistical analysis, and technical writing. Teaching: Surveying Lab, Highway Materials Lab, and Engineering Statics.

Teaching

CIVE 202, Civil Engineering Analysis II 2024 – Present
University of Nebraska-Lincoln, Lincoln, NE, USA

A project-based course teaching civil engineers to program in Python, enabling them to collect, organize, analyze, and visualize data, and automate engineering calculations. Pairs core concepts from mechanics, dynamics, and sustainable design with professional programming skills, so students learn to translate a client's problem into a working solution. Led weekly labs, project debugging, and grading.

Student evaluations (Spring 2024, 2025 and 2026, six sections):

- Mean TA ratings of 4.2 to 4.7 out of 5 across all nine teaching items.
- 100% agreed or strongly agreed they felt welcome and respected (mean 4.62) and were invited to be active participants in their learning.
- 97% agreed course activities promoted their learning and that they knew where to go for help.
- Students most often credited hands-on debugging support in labs, availability for questions, prompt email responses, and explanations of why code failed.

CIVE 361, Principles of Transportation Engineering 2024 – Present
University of Nebraska-Lincoln, Lincoln, NE, USA

Introduces transportation engineering through operator, user, and vehicle characteristics, transportation facility design, traffic operations and control, and transportation planning. Draws on Mannering's *Principles of Highway Engineering and Traffic Analysis* alongside real trajectory and traffic datasets from ongoing research. Graded assignments, held office hours, and supported students by email.

Student evaluations (Fall 2024, 4 responses):

- 75% agreed or strongly agreed that feedback on their work was useful for improvement (mean 4.00).
- Students named timely feedback, support, and communication as the most beneficial elements, citing clear assignment corrections and prompt email help.

CE 343, Surveying Laboratory 2012 – 2013
Jordan University of Science and Technology, Irbid, Jordan

A hands-on lab progressing from taping and chain surveying through levels, theodolites, tacheometers, EDM, and the total station, closing with field projects in traversing, contouring, and area measurement by planimeter.

CE 201, Engineering Statics 2012 – 2013
Jordan University of Science and Technology, Irbid, Jordan

A foundational mechanics course spanning vector analysis, statics of particles, equivalent force systems on rigid bodies, equilibrium, supports and reactions, truss analysis by joints and sections, frames, shear and bending moment diagrams, and centroids and moments of inertia of composite areas.

Mentorship

Undergraduate Research Mentorship

Grace Gabrys Jun – Aug 2026
NSF Research Experiences for Undergraduates (REU) Program
University of Nebraska-Lincoln, Lincoln, NE, USA

Supervised research focusing on the geospatial indexing of pharmaceutical deserts in Nebraska and optimization-based analysis of conventional and autonomous delivery options.

Mawunyo Tchamdja Jun – Aug 2025
NSF Research Experiences for Undergraduates (REU) Program
University of Nebraska-Lincoln, Lincoln, NE, USA

Supervised research focusing on autonomous electric mobility systems in Rural Nebraska as part of the program.

Service

Peer Review

- *Transportation Research Part A: Policy and Practice* ×2
- *Scientific Reports* (Springer Nature) ×1
- *KSCE Journal of Civil Engineering* ×1
- *Transportation Research Board (TRB) Annual Meeting* (2027) ×4
- *Discover Civil Engineering* (Springer Nature) ×1
- *Bridging Transportation Researchers (BTR) Conference*, Nos. 5, 6, 7, 8 ×4

Skills

Areas of expertise	Automated Driving, Traffic Microsimulation, Reinforcement Learning, Machine Learning, Statistical Analysis, Optimization, Traffic Safety Analysis, Crash Analysis, Behavior Heterogeneity, Transport Modeling, Demand Forecast, Traffic Analysis, Systems Accessibility, Soft Mobility, Transport Integration, Transport Planning, Geometric Design
Languages	Python, Julia, R, SPSS, SAS
Simulation & planning	SUMO, VISSIM, VISUM, Synchro, SIDRA, HCS
GIS	ArcGIS, QGIS, GeoDa
Other	MS Office, LaTeX, HTML, Photoshop, Lightroom

References

Name	*Role*	*Organization*	*E-mail*
Wissam Kontar	Assistant Professor	University of Nebraska-Lincoln	wkontar2@nebraska.edu
Jason Hawkins	Assistant Professor	University of Calgary	jfhawkin@ucalgary.ca
Ghazi Al-Khateeb	Professor / Chairman	University of Sharjah	galkhateeb@sharjah.ac.ae
Mohammad Braiwish	Managing Director	TrafQuest	m.younis@trafquest.com